

Decoder-only conditional latent models with temporal mixture priors enable zero-shot cross-lab mouse sleep staging and substage discovery

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Abstract

Manual scoring of mouse sleep from electroencephalography (EEG) and electromyography (EMG) lacks consensus across laboratories, and supervised deep learning models trained at one site often fail on data from others until retrained on multi-laboratory labels. We present an unsupervised framework for population-level sleep staging on the Mouse Sleep Staging Validation dataset (MSSV; OpenNeuro ds006366). A controlled model ladder on identical spectral inputs compares decoder-only conditional Gaussian mixture variational autoencoder (cGMVAE) with decoder-only conditional HMM-GMVAE (cHMM-GMVAE) using a warm sticky HMM-Gaussian mixture model (GMM) latent prior. Models train on leave-mice-out folds pooling laboratories 2, 3, and 5; held-out mice are staged by zero-shot inference—encoding without subject identifiers and assigning states from the learned prior only, without per-mouse manual calibration. On joint cross-lab holdout, cHMM-GMVAE reaches mean best-of-three prior normalised mutual information (NMI) 0.56 (range 0.38–0.65) versus 0.33 for matched cGMVAE ($\Delta = +0.22$ on every fold). Within-laboratory holdout shows site-specific gains: temporal priors help most on laboratories 3 and 5, while static mixture decoders remain competitive on laboratory 2. A substage resolution sweep selects $K=4$ states whose spectra, θ/δ ratios, bout lengths, and transition structure align with Wake, NREM, a transition-enriched state, and REM. Decoder-only conditioning with temporal mixture priors thus improves unsupervised generalisation across mice and acquisition sites while supporting biologically interpretable substage discovery.

Author summary

We study how *Mus musculus* sleep can be classified automatically from brain and muscle signals when recordings come from different laboratories with different equipment. Expert human scorers disagree across labs, and supervised computer models often fail on new sites. We built a family of unsupervised models that learn shared sleep patterns from many mice, then test them on held-out animals they have never seen—without asking a human to calibrate each new recording. Our best model combines a neural network that reads EEG and EMG with a temporal prior that enforces realistic sleep bout structure. It outperforms simpler models on cross-lab holdout and discovers finer sleep substates whose brain-wave profiles match known physiology. We validate substages with power spectra, muscle activity, and bout lengths,

with guidance from sleep neuroscientists. This work offers a label-free route to comparing sleep across studies and may help researchers find transition states between wake, NREM, and REM that are hard to score by eye alone.

Introduction

Rodent sleep is scored from EEG and EMG into wake, non-rapid eye movement (NREM), and rapid eye movement (REM), yet montage and laboratory protocols differ across studies [1]. Rose et al. show that manual scoring lacks cross-lab consensus and that published deep models generalise poorly across MSSV sites until retrained on multi-laboratory data [2]. This motivates label-agnostic staging evaluated under strict population holdout rather than per-animal calibration. Three methodological families address complementary gaps. **Supervised ceilings** such as SPINDLE and REST achieve high cross-lab accuracy but require expert labels that inherit rater noise [3,4]. **Transductive unsupervised** tools including AccuSleep and SegWay fit or calibrate each new recording individually—for example, AccuSleep requires roughly ten minutes of manually scored baseline per mouse [5,6]. **Static unsupervised** methods including FASTER, FASTER2, mcRBM, and WUCSS discover clusters or substates without deep latent sequence priors on multi-laboratory holdout protocols [7–10]. None combine decoder-only population training, sticky HMM–GMM temporal priors, and zero-shot cross-laboratory evaluation on MSSV.

Our MSc thesis established that subject-conditioned GMVAEs reach population NMI ≈ 0.70 on a curated ten-mouse, single-laboratory leave-one-subject-out protocol with encoder+decoder conditioning [11]. **This paper** extends that line with decoder-only conditioning on 92 mice from laboratories 2, 3, and 5, zero-shot holdout inference without subject identifiers at staging, and end-to-end warm HMM–GMM priors [12–15]. We contribute: (i) decoder-only cVAE training with zero-shot holdout staging; (ii) a locked fair comparison of cGMVAE versus cHMM–GMVAE on identical manifests; and (iii) substage biology after a resolution sweep with physiology and transition readouts [9,16]. Dataset, protocol, and model summaries are in Data and Methods; full prior derivations, ablations, and the thesis $K=13$ taxonomy are in the Appendix (Full HMM–GMM prior and warm-start, S6 Fig).

Data

Dataset. All experiments used the Mouse Sleep Staging Validation dataset (MSSV, OpenNeuro ds006366) [1]. Recordings comprise invasive EEG and neck EMG from 92 mice acquired across five European laboratories under a shared protocol, with expert annotations on a fixed 4 s grid for Wake, NREM, REM, and Artifact. This study analyses laboratories 2, 3, and 5, which provide the cross-site variability required for population holdout while avoiding pathology-specific cohorts excluded in the thesis definition [11]. Expert labels are used for evaluation only and are not supplied to the unsupervised objectives. All data were previously published. No new animal procedures were performed.

Epoch construction and preprocessing. Continuous recordings were segmented into 4 s epochs at 128 Hz sampling rate; Artifact epochs were removed before modelling. Per-laboratory spectral front-end settings were fixed from incohort ablation winners (laboratory-specific channel sets and band limits; full locks in S4 Appendix). Montage and acquisition site are treated as nuisance factors controlled by these fixed locks rather than per-mouse manual calibration.

Input representation. Each epoch is a log-power spectral map with channels $c \in \{1, \dots, C\}$ and frequency bins f , giving tensors $\mathbf{x}_t \in \mathbb{R}^{C \times F}$ batched into contiguous sequences for temporal models. The deep ladder learns staging directly from these spectral inputs rather than hand-crafted expert features.

Methods

Cross-validation and holdout protocol

Evaluation scopes. Mice were partitioned into four folds with leave-mice-out cross-validation. Three folds define the training pool and one fold defines held-out validation mice that do not appear in training. Joint scope pools all training-fold mice from laboratories 2, 3, and 5 into one population model and tests on the held-out fold across sites. Within-lab scope restricts both training and validation to a single laboratory while using the same fold splits. Absolute prior NMI values between joint and within-lab scopes are not compared because the training distributions differ.

Holdout inference. Held-out mice are staged without subject identifiers and without decoder reconstruction (Fig 1B). The encoder produces deterministic means μ_t from spectral inputs only; subject embeddings and the decoder are not used. State labels are assigned from the trained GMM or sticky HMM-GMM prior on μ_t (Eq. (4); prior details in Full HMM-GMM prior and warm-start). Per-laboratory spectral locks remain fixed at holdout.

Agreement metrics. Agreement with expert Wake, NREM, and REM labels was quantified using normalised mutual information (NMI) [11]. For partitions Y and $\hat{Y}$,

$$\text{NMI}(Y, \hat{Y}) = \frac{2I(Y; \hat{Y})}{H(Y) + H(\hat{Y})}, \quad (1)$$

where I denotes mutual information and H Shannon entropy. Reported holdout metrics use the maximum prior NMI across three random seeds for each fold and model configuration.

Fig 1. Study overview and zero-shot holdout protocol. **A**, MSSV multi-laboratory cohort with four-fold cross-validation; mice from three folds train a population model evaluated on the held-out fourth fold. **B**, Training uses decoder-only subject conditioning for reconstruction; holdout staging encodes EEG/EMG without subject identifiers and assigns states from the HMM-GMM prior only. **C**, Model ladder: cGMVAE (static GMM prior) and cHMM-GMVAE (sticky HMM-GMM prior); extended baselines in S1 Fig.

Model ladder and architecture

Model ladder. The primary comparison pairs decoder-only cGMVAE (static GMM prior, $T=1$) with decoder-only cHMM-GMVAE (warm sticky HMM-GMM prior, $T=64$) on identical manifests, checkpoints, and $K=3$ mixture components for macro staging (Fig 1C). Both conditional models use subject embeddings in the decoder only during training; HMMGMVAE without conditioning serves as a sequence-prior baseline in S1 Fig [15]. Locked recipes fix latent dimension $d=6$, embedding dimension 8, learning rate 1.3×10^{-3} , 200 epochs, batch size 128, and three random seeds; full hyperparameters are in S1 File.

Encoder-decoder. Training batches consist of contiguous windows $\mathbf{x}_{1:T} \in \mathbb{R}^{T \times C \times F}$ with $T = 64$ epochs per sequence. A convolutional encoder maps each epoch independently to a diagonal Gaussian posterior,

$$q_\phi(\mathbf{z}_t | \mathbf{x}_t) = \mathcal{N}(\mathbf{z}_t | \mu_t, \text{diag}(\sigma_t^2)), \quad (\mu_t, \log \sigma_t^2) = f_\phi(\mathbf{x}_t), \quad (2)$$

where f_ϕ is the CNN-MLP encoder and $\mathbf{z}_t \in \mathbb{R}^d$ with $d = 6$. Latent samples use the reparameterisation $\mathbf{z}_t = \mu_t + \sigma_t \odot \epsilon_t$, $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The decoder reconstructs each epoch from $\mathbf{z}_t$ and, during training only, a subject embedding $\mathbf{e}_i \in \mathbb{R}^8$ for mouse i :

$$\hat{\mathbf{x}}_t = g_\theta(\mathbf{z}_t, \mathbf{e}_i). \quad (3)$$

Under decoder-only conditioning, f_ϕ does not receive $\mathbf{e}_i$; only g_θ does [11].

Variational objective and prior. Training minimises reconstruction error plus a β -weighted KL term (β ramps from 0.01 to 1.0). cGMVAE uses a static Gaussian mixture prior; cHMM-GMVAE replaces it with a sticky HMM-GMM prior ($\kappa=0.92$) warm-started from encoder means (Full HMM-GMM prior and warm-start) [14, 15].

Prior-based staging at holdout. At validation, emissions are evaluated on encoder means μ_t rather than sampled $\mathbf{z}_t$. Static cGMVAE assigns $\hat{s}_t = \arg \max_k [\log \pi_k + \log b_k(\mu_t)]$. cHMM-GMVAE applies Viterbi decoding on $B_{tk} = \log b_k(\mu_t)$,

$$\hat{\mathbf{s}}_{1:T} = \arg \max_{\mathbf{s}_{1:T}} \left[\log \pi_{s_1} + \sum_{t=2}^T \log A_{s_{t-1}, s_t} + \sum_{t=1}^T \log b_{s_t}(\mu_t) \right], \quad (4)$$

contrasting with transductive per-animal calibration [5, 6].

Training, substages, and reproducibility

Decoder-only conditioning. Subject identity enters only the decoder path (Eq. (3)), not the encoder (Eq. (2)), unlike the thesis encoder+decoder setup [11]. Nuisance amplitude and montage effects are absorbed during reconstruction while encoder means remain free of identity information for holdout staging.

Training and checkpoints. Models trained on all mice in the training folds; validation prior NMI monitored each epoch and the best checkpoint per seed retained. Macro ladder comparisons use $K=3$ components; a substage sweep varies $K \in \{3, 4, \dots, 15\}$ on joint holdout fold 4. We selected $K=4$ as the highest K within one standard deviation of the peak mean NMI with non-degenerate occupancy across seeds (S2 Fig). Physiology summaries use EEG power spectra, θ/δ ratio, bout length, expert macro composition, and row-normalised transition matrices [9].

Data and code availability. MSSV is available on OpenNeuro (ds006366). Processed features, fold manifests, and locked configuration files are in the SPA repository (S1 File). Training, evaluation, and figure generation code are maintained in the SPA repository; a version-tagged Zenodo archive will be released before journal submission.

Results and discussion

Cross-lab holdout performance

We evaluated decoder-only conditional models under joint population training with leave-mice-out four-fold cross-validation across MSSV laboratories 2, 3, and 5. Training uses subject embeddings in the decoder only; holdout staging assigns states from the learned prior without subject identifiers (Fig 1B), contrasting with per-mouse calibration in AccuSleep [5]. Fig 2 and Table 1 summarise prior NMI on held-out mice.

Fig 2. Joint cross-lab holdout prior NMI. **A**, Mean best-of-three prior NMI across four folds; error bars span fold min-max; bracket shows mean $\Delta(\text{cHMM-cGMVAE}) = +0.22$. **B**, Per-fold grouped bars (laboratories 2, 3, and 5 pooled). Random three-class assignment yields NMI ≈ 0 .

Table 1. Joint holdout prior NMI by fold (best-of-three seeds, locked recipes).

Fold	cGMVAE	cHMM-GMVAE
1	0.361	0.612
2	0.275	0.381
3	0.407	0.594
4	0.296	0.646
Mean	0.335	0.558

In Table 1, cHMM-GMVAE exceeds cGMVAE on every fold (mean prior NMI 0.56 vs. 0.33). Compared with chance-level macro agreement, both models extract structure, but the temporal prior adds a consistent margin ($\Delta = +0.22$ mean). The gain was largest on folds 1 and 4 (+0.25 and +0.35 NMI) and smallest on fold 2 (+0.11), which mixes difficult laboratory-2 montages with held-out mice from all sites. Per-mouse fold-4 analysis (S4 Fig) shows the improvement is distributed across animals rather than driven by a single outlier. Macro confusion on fold 4 (S3 Fig) confirms strong Wake and NREM alignment for the $K=3$ macro model. Rose et al. establish that cross-lab supervised staging inherits rater disagreement [2]; our unsupervised ladder discovers shared structure without fitting one laboratory’s label biases. Extended ladder rungs including HMMGMVAE appear in S1 Fig.

Laboratory heterogeneity

Fig 3 summarises within-laboratory holdout: mean prior NMI by laboratory and model. cHMM-GMVAE leads on laboratories 3 and 5 (mean gains +0.05 and +0.06 NMI over cGMVAE), whereas laboratory 2 shows near parity (+0.04). Mechanistically, sticky sequence priors help when bout structure and spectral signatures

align with temporal smoothing—as on the longer, more regular NREM bouts typical of laboratories 3 and 5—while a static GMM decoder already captures laboratory-2 montages adequately. This heterogeneity is expected under fixed per-laboratory spectral locks that do not adapt to individual held-out mice. Colleague work on the same MSSV cohort (QSLP-AE) targets cross-species representation alignment rather than cross-laboratory zero-shot holdout [17]; the problems are complementary. Full within-lab fold tables are in S2 Table.

Fig 3. Laboratory heterogeneity under within-lab holdout. **A**, Mean prior NMI (four folds) for cGMVAE and cHMM-GMVAE by laboratory; numbers in cells; right column: $\Delta(\text{cHMM}-\text{cGMVAE})$. Absolute values are not compared to joint-scope NMI in Fig 2.

Substage resolution and physiology

Macro ladder comparisons use $K=3$. On joint holdout fold 4 we swept $K \in \{3, 4, \dots, 15\}$ (S2 Fig): mean validation NMI peaks at $K=4$ (best-of-three 0.643; $K=3$ at 0.635), with sharp declines beyond $K=6$, indicating over-segmentation. Fig 4 shows compact physiology for the $K=4$ winner (seed 3). Wake and NREM states separate by θ/δ and bout length; a transition-enriched state shows mixed expert macro composition at Wake–NREM boundaries; the fourth state aligns with REM spectra. This extends mcRBM-style substage discovery [9] under a population holdout protocol rather than pooled single-lab fitting. Finer thesis taxonomies ($K=13$) appear in S5 Fig; expert-curated names are in S3 Table.

Fig 4. Substage physiology at $K=4$ (fold 4 holdout, seed 3). **A**, Mean EEG power spectra per discovered state. **B**, θ/δ ratio (NREM depth proxy). **C**, Bout length distributions. **D**, Expert macro composition per state. Provisional state names derive from dominant expert macro labels.

Dynamics and transitions

Fig 5 links substates to temporal structure. The row-normalised transition matrix highlights off-diagonal flow between Wake and NREM-dominated states and a transition-enriched state, consistent with supervised transition classifiers [16, 18]. The two-hour hypnogram excerpt shows inferred substates tracking expert macro epochs while resolving brief Wake–NREM transitions that collapse to single macro labels. Together with Fig 4, the dynamics support interpreting $K=4$ as a biologically meaningful refinement rather than arbitrary cluster count.

Fig 5. Substage dynamics ($K=4$, fold 4 holdout). **A**, Row-normalised transition matrix (probability of next state given a state change). **B**, Two-hour hypnogram: discovered substates (top) and expert macro labels (bottom).

Limitations

Agreement is quantified by NMI against expert macro labels, which inherit cross-laboratory disagreement [2]; high NMI therefore measures discovery of shared structure, not mimicry of a single rater. Absolute holdout NMI (0.56 mean) remains below thesis leave-one-subject-out performance on a curated single-lab subset because cross-laboratory shift is harder than within-lab LOSO [11]. Seed and fold variance (fold 2) indicate remaining fragility. Fixed spectral front-end locks control laboratory nuisance but do not replace per-mouse calibration; the trade-off is intentional zero-shot staging without manual baseline scoring. Decoder-only conditioning is the main advance over thesis encoder+decoder conditioning for population holdout; ablation evidence is in S3 Appendix.

Conclusion

Decoder-only conditional HMM–GMVAE with sticky mixture priors improves zero-shot cross-laboratory mouse sleep staging on MSSV relative to matched cGMVAE (mean prior NMI 0.56 vs. 0.33). Subject embeddings at training explain nuisance variability without leaking identity at holdout inference; temporal priors capture bout structure where laboratory montages support sequence modelling. Substage physiology and transition analyses at $K=4$ extend macro staging toward interpretable Wake, NREM, transition, and REM structure under rigorous population holdout.

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Appendix

Additional holdout results

S1 Fig. Per-fold holdout ladder. Grouped bar chart of prior NMI for all four folds; includes HMMGMVAE baseline when available.

S2 Fig. Substage resolution sweep. Validation NMI and predictive log-likelihood versus K on locked cHMM–GMVAE (joint fold 4); shaded band: seed min–max; dashed line: chosen $K=4$.

S3 Fig. Macro confusion (fold 4, $K=3$). Row-normalised expert macro composition per predicted state (A) and macro recall (B).

S4 Fig. Per-mouse prior NMI on fold 4. Paired bars for cGMVAE and cHMM–GMVAE by held-out mouse.

S5 Fig. Full substage physiology grid ($K=4$). Seven-column thesis-style panel (EEG/EMG PSD, features, bout length, macro and subject mixing).

S2 Table. Within-lab holdout prior NMI. Full table from holdout ladder summary.

S3 Table. Substage summary. Dominant macro label, bout length, θ/δ , and expert names after curation.

Full HMM–GMM prior and warm-start

Each discrete latent state $s_t \in \{1, \dots, K\}$ emits a diagonal Gaussian in $\mathbb{R}^d$. The per-step emission log-density is

$$\log b_k(\mathbf{z}_t) = \log \mathcal{N}(\mathbf{z}_t \mid \mathbf{m}_k, \text{diag}(\mathbf{v}_k)). \quad (5)$$

Transitions are initialised sticky with $\kappa = 0.92$. The marginal log-likelihood of a latent window is

$$\log p(\mathbf{z}_{1:T}) = \log \sum_{s_{1:T}} \pi_{s_1} \prod_{t=2}^T A_{s_{t-1}, s_t} \prod_{t=1}^T b_{s_t}(\mathbf{z}_t), \quad (6)$$

and the sequence KL averages

$$\mathcal{L}_{\text{KL}} = \frac{1}{T} \sum_{t=1}^T [\log q_\phi(\mathbf{z}_t \mid \mathbf{x}_t) - \log p(\mathbf{z}_{1:T})]. \quad (7)$$

Warm-start schedule: epochs 1–17 use standard Gaussian KL; epoch 18 initialises GMM emissions from K -means on encoder means; epochs 18–36 use independent mixture KL; from epoch 37 the HMM–GMM prior blends in over 18 ramp epochs [14, 15]. CNN architecture: channel widths [32, 64, 128, 128], kernel sizes [7, 5, 5, 3], strides 2, MLP widths [512, 256, 128].

Ablations and validation

S2 Appendix. Raw time-domain HMM failure. Population NMI near zero on MSSV raw EEG/EMG (thesis Ch. 5).

S3 Appendix. Decoder-only vs encoder+decoder ablation. Core novelty versus thesis conditioning.

S4 Appendix. Preprocessing locks. Per-laboratory channel sets and band limits (laboratory 2: EEG 0–30 Hz; laboratories 3/5: 0–20 Hz).

S1 Appendix (synthetic). Synthetic HMM validation. Optimisation and latent recovery benchmarks (thesis Ch. 3).

Configuration and literature

S1 File. Locked YAML recipes. Joint and within-lab holdout configuration files in the SPA repository.

S1 Table. Extended literature comparison. Supervised ceilings, transductive unsupervised, and static unsupervised methods versus this work.

S6 Fig. Thesis cGMVAE $K=13$ taxonomy. Full PCA, transition graph, and panel from the MSc thesis [11].

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